

Global Superstore Dataset Report

Data Quality Analysis, Transformation Validation and Exploratory Analysis

Dataset Source: Kaggle - Global Superstore

Report Purpose: To analyse the structure, quality, transformations and important characteristics of the cleaned dataset.

1. Dataset Overview

Metric	Value
Number of Records	51,290
Number of Columns	25
Duplicate Rows	0
Duplicate Row IDs	0
Countries	147
Markets	7
Regions	13
Product Categories	3
Earliest Order Date	2011-01-01
Latest Order Date	2014-12-31

The dataset contains **51,290 records** and **25 columns**. It contains information about customers, products, locations, sales, profits, shipping and order priority.

2. Column Information

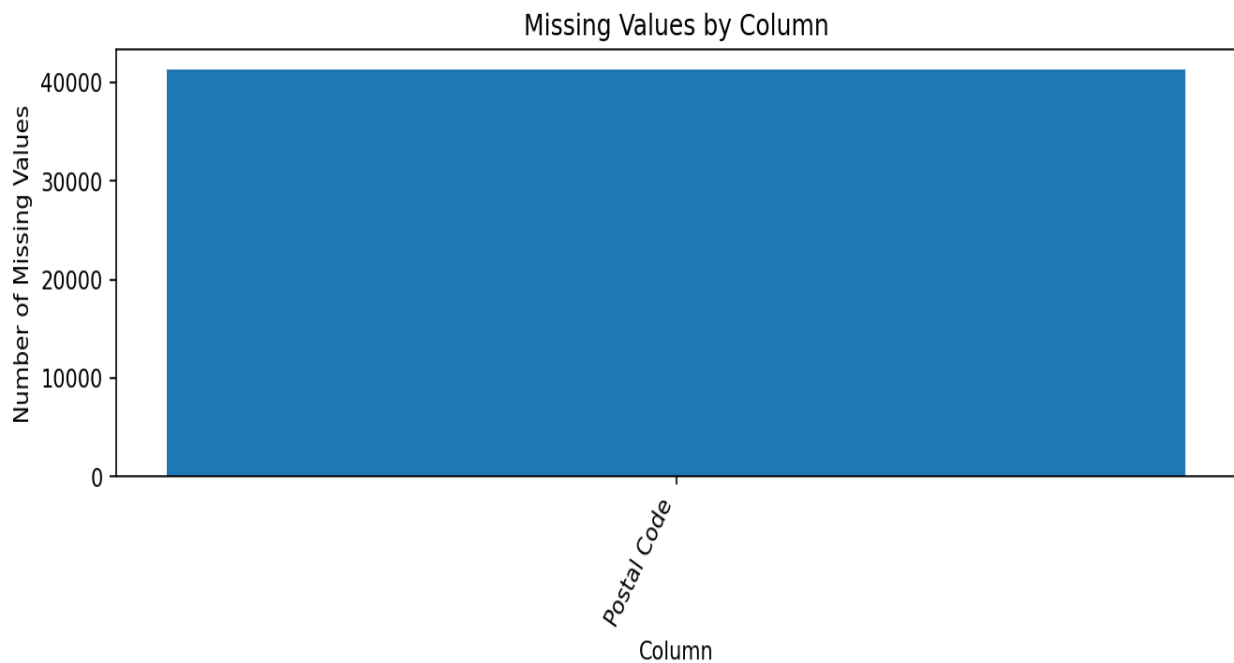
Column	Data Type	Non-Null	Missing	Unique
Row ID	int64	51,290	0	51,290
Order ID	object	51,290	0	25,035
Order Date	datetime64[ns]	51,290	0	1,430
Ship Date	datetime64[ns]	51,290	0	1,464
Ship Mode	object	51,290	0	4
Customer ID	object	51,290	0	1,590
Customer Name	object	51,290	0	795
Segment	object	51,290	0	3
City	object	51,290	0	3,636
State	object	51,290	0	1,094
Country	object	51,290	0	147
Postal Code	float64	9,994	41,296	631
Market	object	51,290	0	7
Region	object	51,290	0	13
Product ID	object	51,290	0	10,292
Category	object	51,290	0	3
Sub-Category	object	51,290	0	17
Product Name	object	51,290	0	3,788
Sales	float64	51,290	0	22,995
Quantity	int64	51,290	0	14
Discount	float64	51,290	0	27

Column	Data Type	Non-Null	Missing	Unique
Profit	float64	51,290	0	24,575
Shipping Cost	float64	51,290	0	10,037
Order Priority	object	51,290	0	4
Shipping Days	int64	51,290	0	8

3. Missing Value Analysis

Column	Missing Values	Missing %
Postal Code	41,296	80.51
Order ID	0	0.00
Order Date	0	0.00
Ship Date	0	0.00
Ship Mode	0	0.00
Customer ID	0	0.00
Customer Name	0	0.00
Segment	0	0.00
Row ID	0	0.00
City	0	0.00
State	0	0.00
Country	0	0.00
Market	0	0.00
Region	0	0.00
Product ID	0	0.00
Category	0	0.00
Sub-Category	0	0.00
Product Name	0	0.00
Sales	0	0.00
Quantity	0	0.00
Discount	0	0.00
Profit	0	0.00
Shipping Cost	0	0.00
Order Priority	0	0.00
Shipping Days	0	0.00

The main missing-value issue is the **Postal Code** column, which contains **41,296** missing values. These values were retained because reliable postal codes cannot be safely inferred from the available data.



4. Duplicate Analysis

Check	Count
Duplicate complete rows	0
Duplicate Row IDs	0

No duplicate records were identified. Row ID values were also checked and found to be unique.

5. Numerical Data Analysis

Variable	count	mean	std	min	25%	50%	75%	max	Median	Missing
Sales	51,290.00	246.49	487.57	0.44	30.76	85.05	251.05	22,638.48	85.05	0
Quantity	51,290.00	3.48	2.28	1.00	2.00	3.00	5.00	14.00	3.00	0
Discount	51,290.00	0.14	0.21	0.00	0.00	0.00	0.20	0.85	0.00	0
Profit	51,290.00	28.61	174.34	-6,599.98	0.00	9.24	36.81	8,399.98	9.24	0
Shipping Cost	51,290.00	26.38	57.30	0.00	2.61	7.79	24.45	933.57	7.79	0
Shipping Days	51,290.00	3.97	1.73	0.00	3.00	4.00	5.00	7.00	4.00	0

The numerical attributes were inspected for unusual or invalid values. Negative Profit values were retained because they represent legitimate business losses rather than data-entry errors.

6. Categorical Data Analysis

Ship Mode

Value	Count	Percentage
Standard Class	30,775	60.00
Second Class	10,309	20.10
First Class	7,505	14.63
Same Day	2,701	5.27

Segment

Value	Count	Percentage
Consumer	26,518	51.70
Corporate	15,429	30.08
Home Office	9,343	18.22

Category

Value	Count	Percentage
Office Supplies	31,273	60.97
Technology	10,141	19.77
Furniture	9,876	19.26

Sub-Category

Value	Count	Percentage
Binders	6,152	11.99
Storage	5,059	9.86
Art	4,883	9.52
Paper	3,538	6.90
Chairs	3,434	6.70
Phones	3,357	6.55
Furnishings	3,170	6.18
Accessories	3,075	6.00
Labels	2,606	5.08
Envelopes	2,435	4.75
Supplies	2,425	4.73
Fasteners	2,420	4.72
Bookcases	2,411	4.70
Copiers	2,223	4.33
Appliances	1,755	3.42

Value	Count	Percentage
Machines	1,486	2.90
Tables	861	1.68

Market

Value	Count	Percentage
APAC	11,002	21.45
LATAM	10,294	20.07
EU	10,000	19.50
US	9,994	19.49
EMEA	5,029	9.81
Africa	4,587	8.94
Canada	384	0.75

Region

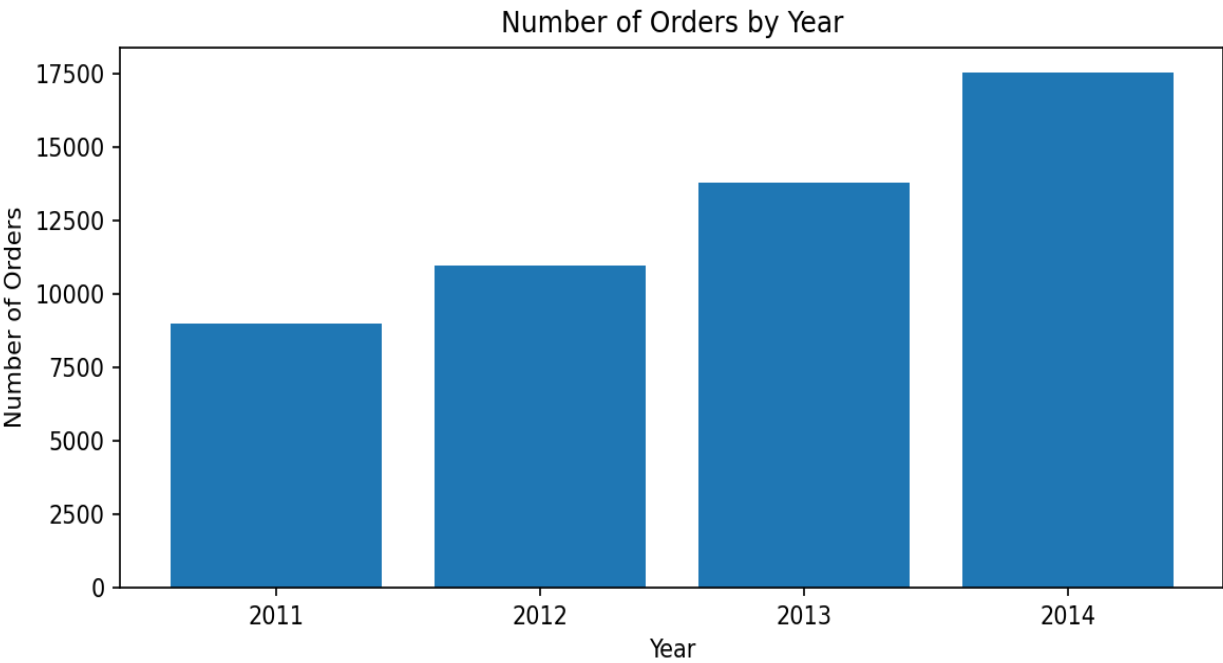
Value	Count	Percentage
Central	11,117	21.67
South	6,645	12.96
EMEA	5,029	9.81
North	4,785	9.33
Africa	4,587	8.94
Oceania	3,487	6.80
West	3,203	6.24
Southeast Asia	3,129	6.10
East	2,848	5.55
North Asia	2,338	4.56
Central Asia	2,048	3.99
Caribbean	1,690	3.29
Canada	384	0.75

Order Priority

Value	Count	Percentage
Medium	29,433	57.39
High	15,501	30.22
Critical	3,932	7.67
Low	2,424	4.73

7. Date Analysis

Metric	Value
Earliest Order Date	2011-01-01
Latest Order Date	2014-12-31
Number of Years	4
Number of Months	48
Invalid Shipping Dates	0



8. Data Transformation Summary

Transformation / Check	Result
Order Date conversion	Converted to datetime
Ship Date conversion	Converted to datetime
Duplicate row check	No duplicates found
Duplicate Row ID check	No duplicates found
Invalid date check	No invalid dates found
Missing Postal Code handling	Missing values retained as NULL
Numerical data validation	Values checked
Categorical data validation	No obvious inconsistencies found
Negative Profit handling	Retained as valid business losses
Shipping Days creation	Ship Date - Order Date

The transformation stage focused on improving data consistency and validating the dataset without unnecessarily removing valid records.

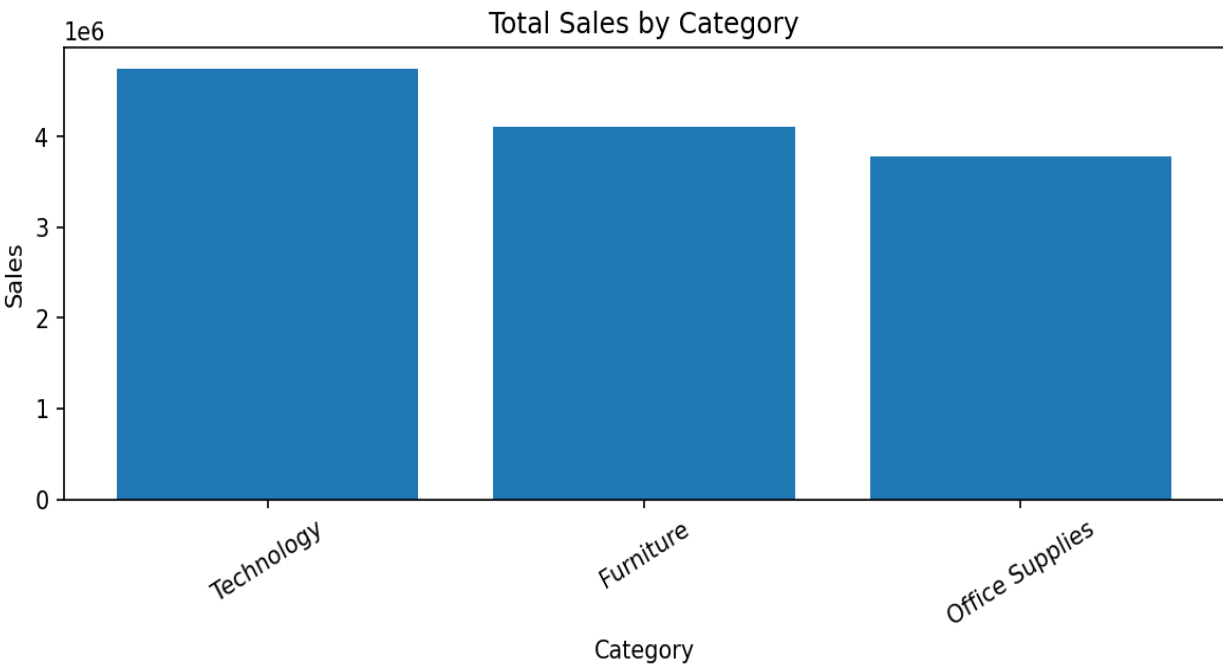
9. Shipping Days Analysis

index	count	mean	std	min	25%	50%	75%	max
Shipping Days	51,290.00	3.97	1.73	0.00	3.00	4.00	5.00	7.00



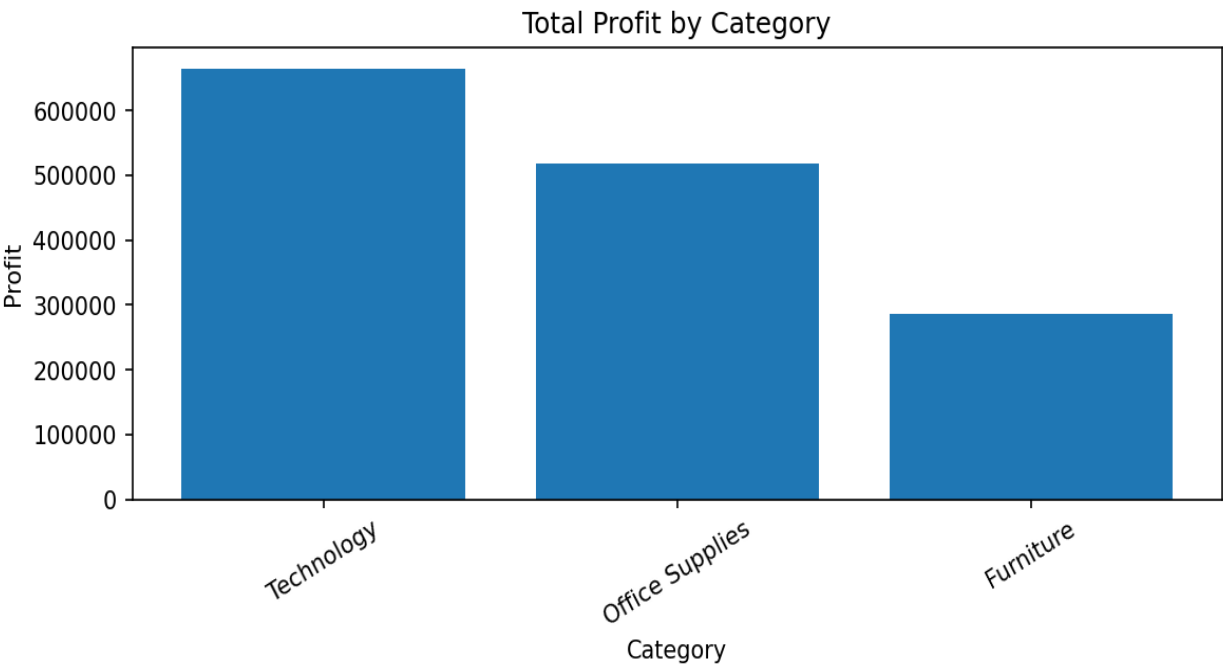
10. Sales by Category

Category	Total Sales
Technology	4,744,557.50
Furniture	4,110,874.19
Office Supplies	3,787,070.23



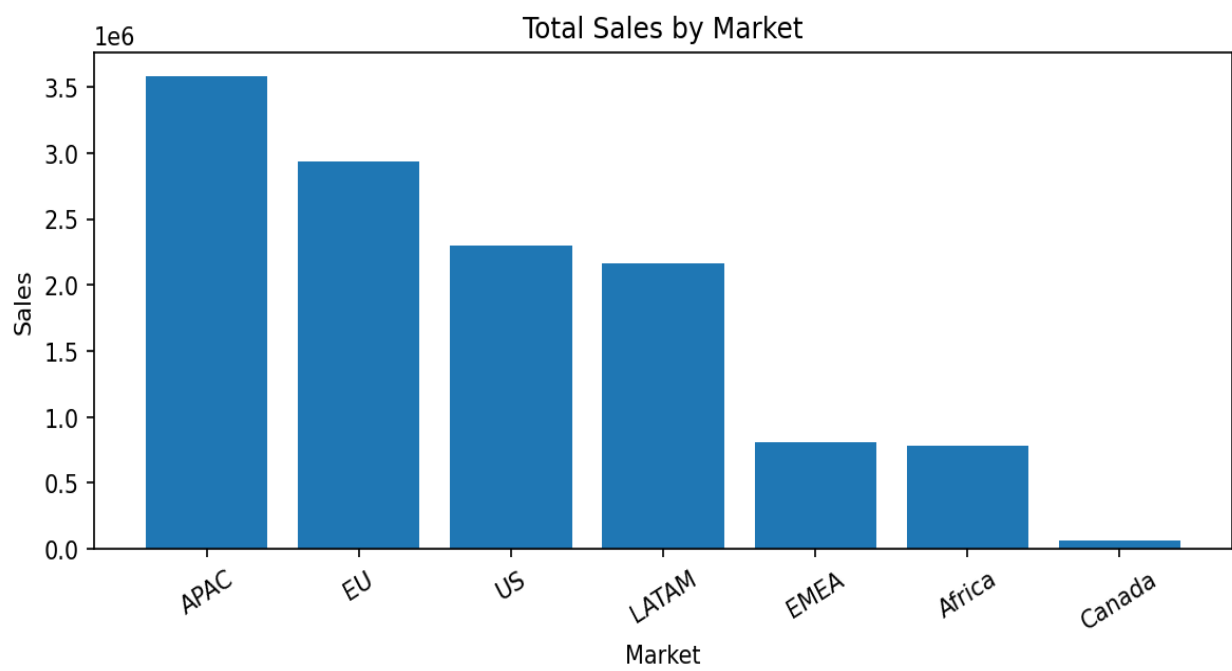
11. Profit by Category

Category	Total Profit
Technology	663,778.73
Office Supplies	518,473.83
Furniture	285,204.72



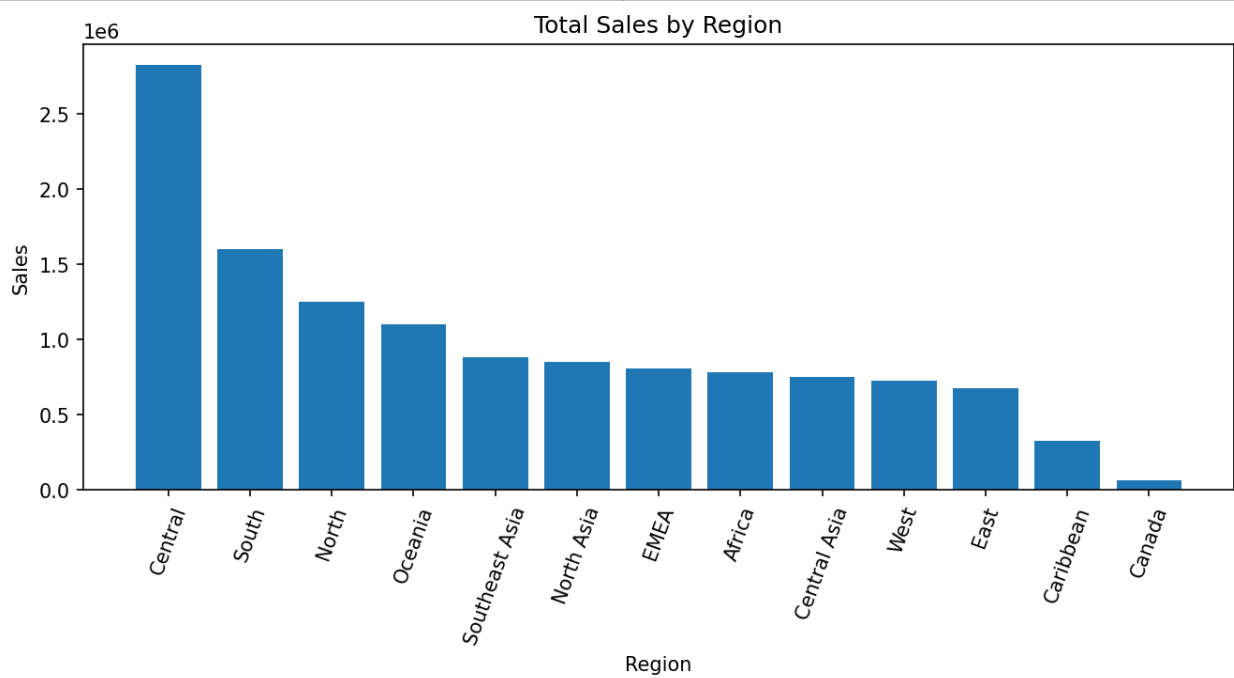
12. Sales by Market

Market	Total Sales
APAC	3,585,744.13
EU	2,938,089.06
US	2,297,200.86
LATAM	2,164,605.17
EMEA	806,161.31
Africa	783,773.21
Canada	66,928.17



13. Sales by Region

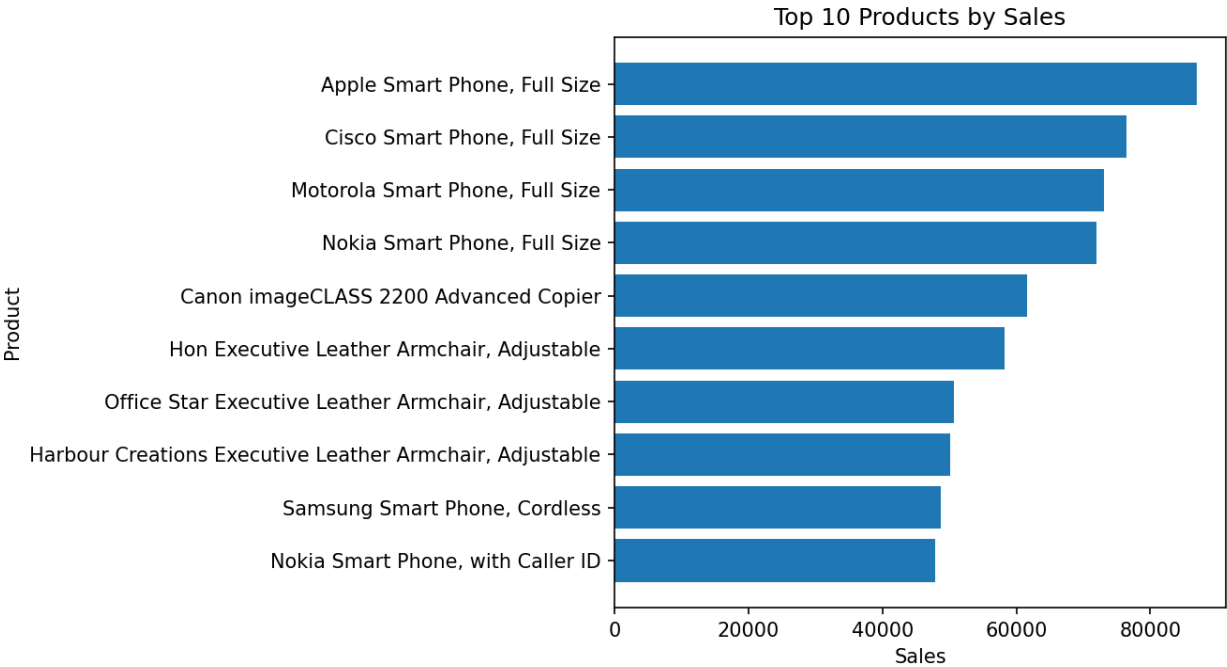
Region	Total Sales
Central	2,822,302.52
South	1,600,907.04
North	1,248,165.60
Oceania	1,100,184.61
Southeast Asia	884,423.17
North Asia	848,309.78
EMEA	806,161.31
Africa	783,773.21
Central Asia	752,826.57
West	725,457.82
East	678,781.24
Caribbean	324,280.86
Canada	66,928.17



14. Top 10 Products by Sales

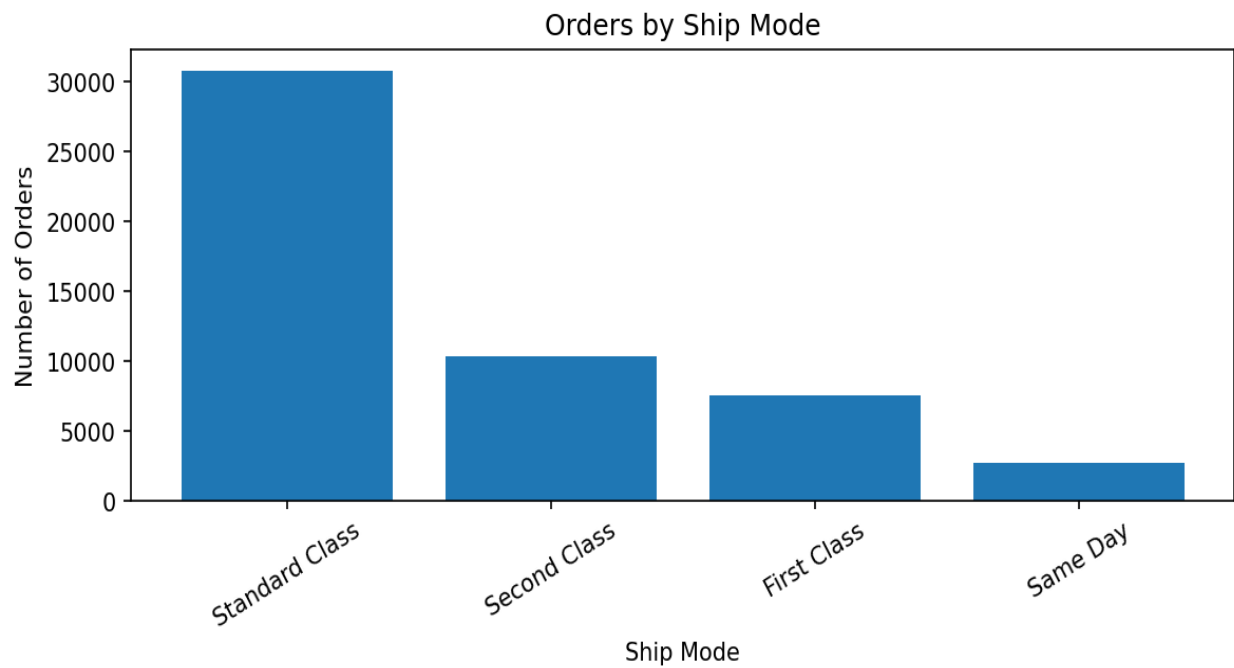
Product Name	Total Sales
Apple Smart Phone, Full Size	86,935.78
Cisco Smart Phone, Full Size	76,441.53
Motorola Smart Phone, Full Size	73,156.30
Nokia Smart Phone, Full Size	71,904.56
Canon imageCLASS 2200 Advanced Copier	61,599.82
Hon Executive Leather Armchair, Adjustable	58,193.48

Product Name	Total Sales
Office Star Executive Leather Armchair, Adjustable	50,661.68
Harbour Creations Executive Leather Armchair, Adjustable	50,121.52
Samsung Smart Phone, Cordless	48,653.46
Nokia Smart Phone, with Caller ID	47,877.79



15. Ship Mode Analysis

Ship Mode	Number of Orders
Standard Class	30,775
Second Class	10,309
First Class	7,505
Same Day	2,701



16. Order Priority Analysis

Order Priority	Number of Orders
Medium	29,433
High	15,501
Critical	3,932
Low	2,424

17. Key Statistics

Metric	Value
Total Sales	12,642,501.91
Total Profit	1,467,457.29
Total Quantity Sold	178,312
Average Sales per Record	246.49
Average Profit per Record	28.61
Average Shipping Days	3.97

18. Key Findings

1. The dataset contains **51,290 records** and **25 columns**.
2. No duplicate rows or duplicate Row IDs were identified.
3. The Postal Code column contains **41,296** missing values. These were retained as NULL because they could not be reliably inferred.
4. The **Technology** category generated the highest total sales.
5. The **Technology** category generated the highest total profit.
6. The **APAC** market generated the highest total sales.
7. The **Central** region generated the highest total sales.
8. The most frequently used shipping mode was **Standard Class**.
9. Negative Profit values were retained because they represent legitimate business losses.
10. Shipping Days was created by calculating the difference between Ship Date and Order Date.

19. Final Data Quality Summary

Quality Check	Result
Duplicate Rows	0
Duplicate Row IDs	0
Invalid Shipping Dates	0
Missing Postal Codes	41,296
Missing Shipping Days	0
Records After Transformation	51,290

20. Conclusion

The Global Superstore dataset was successfully extracted and transformed for further analysis and data warehousing.

The dataset was checked for duplicate records, missing values, invalid dates, numerical inconsistencies and categorical inconsistencies. The Order Date and Ship Date fields were converted to appropriate datetime formats.

Missing Postal Code values were retained because reliable values could not be inferred. Negative Profit values were also retained because they represent legitimate business losses.

A new Shipping Days attribute was created using the difference between Ship Date and Order Date.

The analysis provided insights into sales, profit, categories, markets, regions, products, shipping methods and order priorities.

The cleaned dataset is now ready for the Load stage of the ETL pipeline.